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Experience:

- **Sardar Vallabhbhai National Institute of Technology, Surat**
Assistant Professor, October 2023 – Present
- **Gatishakti Vishwavidyalaya, Vadodara**
Visiting Faculty, September 2025 December 2025
- **National Institute of Technology, Warangal**
Ad-hoc Faculty, Aug 2022 to May 2023

Received an invitation for Postdoctoral Research, Chonnam National University, South Korea, April 2023.

Education:

- Doctor of Philosophy (**Ph.D.**) in Chemical Engineering, Visvesvaraya National Institute of Technology (VNIT), Nagpur (2019–2022)
- Master in Technology (**M. Tech**) in Chemical Engineering, UICT, Jalgaon (2016–2018)
- Bachelor of Technology (**B. Tech**) in Chemical Engineering, UICT, Jalgaon (2012–2016)

Research Interests:

- Nanomaterials Applications in Process Intensification and Sustainable Biological Processes
- Data-Driven Sustainable Processes and Environmental Engineering

Professional Summary

Category	Details
Journal Publications	11 (6 SCIE+5 Scopus indexed) 2 Q1 4 Q2 2 Q3 1 Q4 2 Conference Proceedings
Granted Patents	3
Sponsored Projects	Seed Grant ₹10 Lakh
Guest Editor	2 SCIE Journals
Authored Books	2 (CRC Press)
Book Chapters	35 (ELSEVIER)
Google Scholar	930 citations H-index 15
Scopus	710 citations H-index 13
PhD Students	6 (Ongoing)
M.Tech Students	2
Conference Activities	37 (1 Organized 4 Session Chair 9 Invited Talks 20 Paper Presented 3 Conference Attended)
Workshops/STTPs	07 Organized 33 Attended

Institute Administrative Roles:

- Warden: Bhabha Bhawan
- Member Secretary: Students Orientation Programs
- Member: NIRF Committee
- Member: BTech Admission & Fee Remission
- Co-Chairperson: Mind Bend: largest technical event
- Co-Chairperson: ChES & BIS Student Chapters
- Department Coordinator: Career Development Cell, Internships, Media Cell and Website updates

Detailed CV

SECTION A: Teaching and Student Guidance



➤ Courses Taught (Theory)

Sr. No.	Subject	Subject Code	Programme-Year-Semester	AY
1	Programming for Chemical Engineers (Developed Curriculum)	CH 108	BTech-1 st yr-II nd Sem	2024-25 2025-26
2	Engineering Mathematics	CH 202	BTech-2 nd yr-III rd Sem	2023-24
3	Introduction to Macro-Molecules (Developed Curriculum)	CH 252	BTech-2 nd yr-III rd Sem	2024-25 to 2026-27
4	Data Science for Chemical Engineers (Developed Curriculum)	CH 374	BTech-3 rd yr-V th Sem	2025-26 2026-27
5	Nanomaterials Synthesis by Chemical Method	CH 427	BTech-4 th yr-VII th Sem	2024-25

➤ Laboratory Courses

Sr. No.	Subject	Subject Code	Programme-Year-Semester	AY
1	Programming for Chemical Engineers	CH 108	BTech-1 st year-II nd Sem	2024-25 2025-26
2	Fluid flow operations	CH 203	BTech-2 nd year-III rd Sem	2025-26 2026-27
3	Chemical Reaction Engg-I	CH 204	BTech-2 nd year-IV th Sem	2023-24
4	Heat Transfer	CH 205	BTech-2 nd year-III rd Sem	2024-25 to 2026-27
5	Mass Transfer	CH 206	BTech-2 nd year-IV th Sem	2023-24
6	Mass Transfer-II	CH 301	BTech-3 rd year-V th Sem	2024-25
7	Instrumentation & Process Control	CH 302	BTech-3 rd year-VI th Sem	2023-24
8	Chemical Engg -I	CHCH 104	MTech-1 st year-I st Sem	2024-25

➤ Students Guidance

PhD (Ongoing)

Sr No	Name	Roll No.	Admission (Category)	Role	Co-Supervisor/ Supervisor
1	Jayraj Rana	D25CH001	July 2025 (Full-Time)	Supervisor	Dr. Arvind Mungray
2	Jay Narang	D25CH002	July 2025 (Full-Time)	Co-Supervisor	Dr. Arvind Mungray
3	Piyush Laxkar	D26CH002	July 2026 (Full-Time)	Supervisor	Dr. Arvind Mungray
4	Vikas Gupta	D26CH003	July 2026 (Full-Time)	Supervisor	Dr. Arvind Mungray
5	Ashwini Kharche	D26CH007	July 2026 (Part-Time)	Supervisor	Dr. Amrut Mulay
6	Sreshtha Paul	D26CH010	July 2026 (Full-Time)	Supervisor	Dr. Lokesh Ramteke

MTech

Sr. No.	Name	Roll No.	Batch (Passout)	Role	Name of Co-Guide/Guide	Status
1	Raj Parmar	P24CH005	July 2026	Co-Guide	Dr. Arvind Mungray	Completed
2	Nishit Patel	P24CH004	July 2026	Co-Guide	Dr. V. N. Lad	Completed
3	Alok Mishra		July 2027	Co-Guide	Dr. Arvind Mungray	Ongoing

- **BTech Project Guidance:** Guided 9 B Tech Students for Bachelors project. Currently guiding 10 B Tech students
 - **Organized mock interview programs for 100+ B. Tech students (2024, 2025, 2026)**
- **Guest Lecture organized at DoChE, SVNIT, Surat:**
1. Dr. Nirav Lakinwala, Senior Associate, Air Quality Sector, Centre for Study of Science, Technology and Policy (CSTEP), Bengaluru (25 June 2026)
 2. Dr. Bharat Jain, Member Secretary, Gujarat Cleaner Production Centre (GCPC), Gandhinagar (25 March 2026)
 3. Mr. Aashish Mehta, Vice President, Luthra Group, Surat (22 April 2026)
- **One-day Industrial Visit organized for students of DoChE, SVNIT, Surat:**
4. JK Lakshmi Cements Plant, Kadodara, Surat (10 August 2024)
 5. ONGC, Hazira, Surat (16 February 2025)
- **Students Achievement:**
- Research paper by PhD Scholar **Mr. Jayraj Rana accepted for presentation** at the **2nd Global Cleaner Production Conference** at Melia Sitges , Spain organized by Journal of Cleaner Production, ELSEVIER, (Impact Factor 10.7, Q1) 26–29 October 2026.
 - Guided B.Tech student team that received **₹20,000 project funding from ASHINE-SVNIT** for development of Stable Hybrid Nanofluids, 2025-26
 - Mentored B.Tech student team that secured 2nd place at **ASTHRA-A National level Technical Symposium** at Department of Chemical Engg, BVRIT, Narsapur, Hyderabad, Telangana (17-18 Oct, 2025)

SECTION B: Research Activities



➤ Publications in Journals:

1. Jayraj Rana, Parag Thakur, Malika Manjakuppam, Arvind Kumar Mungray, Shriram Sonawane, Group III–V xenes at the biointerface: Monoelemental 2D platforms for biosensing, cancer nanotheranostics and regeneration, Nano-Structures & Nano-Objects, Volume 46, May 2026 (<https://doi.org/10.1016/j.nanoso.2026.101628>); (Q1: Scopus: Impact Factor: NA)
2. Shriram S. Sonawane, Parag P. Thakur, Manjakuppam Malika, Hafiz Muhammad Ali, Recent Advances in the Applications of Green Synthesized Nanoparticle Based Nanofluids for the Environmental Remediation, Current Pharmaceutical Biotechnology, 2023;24(1):188-198 ([10.2174/1389201023666220411114620](https://doi.org/10.2174/1389201023666220411114620)); (Q2: SCIE: Impact Factor:2.83)
3. Parag P. Thakur, Shriram S. Sonawane, Hussein A Mohammed, Recent Trends in Applications of Nanofluids for Effective Utilization of Solar Energy, Current Nano-Science, 2023, vol. 19, issue 2, pp. 170-185; ([10.2174/1573413718666220119104138](https://doi.org/10.2174/1573413718666220119104138)); (Q3: SCIE: Impact Factor:1.51)
4. Parag Thakur, Nishant Kumar, Shriram Sonawane, Enhancement of pool boiling performance using MWCNT based nanofluids: A sustainable alternative for the heat recovery of wastewater and incinerator, Sustainable Energy Technologies and Assessments, Volume 45, 2021, 101-115, (<https://doi.org/10.1016/j.seta.2021.101115>); (Q1: SCIE: Impact Factor:7.63)
5. Parag Thakur, Shriram Sonawane, Irina Potoroko, Shirish H Sonawane, Recent Advances in Ultrasound-assisted Synthesis of Nano-emulsions and their Industrial Applications, Current Pharmaceutical Biotechnology, Volume 22, Issue 13, 2021; 1748 – 1758. (<https://doi.org/10.2174/1389201021666201104150102>); (Q2: SCIE: Impact Factor:2.83)
6. Thakur, P. P., Khapane, T. S., & Sonawane, S. S. Comparative performance evaluation of flyash based hybrid nanofluids in micro-channel based direct absorption solar collector. Journal of Thermal Analysis and Calorimetry, 143, pages 1713–1726 (2021). (<https://doi.org/10.1007/s10973-020-09884-5>); (Q2: SCIE: Impact Factor:4.75)
7. Nirvik Sen, Parag Thakur, Sameer Ekhande, R. Sirsam, K.K. Singh, S. Mukhopadhyaya, K.T.Shenoy, Reactive stripping and precipitation of Uranium in Micro-reactors, Separation Science & Technology, 54(9), 1430-1442, 2019. (<https://doi.org/10.1080/01496395.2018.1563158>) (Q2: SCIE: Impact Factor:2.8)
8. Parag Thakur, Shriram Sonawane, Numeric and experimental investigations of Fe₂O₃ based nanofluids in direct absorption solar collector, Journal of Indian Chemical Society, 2020, 97 (10a), 1636-1641. I.F.- 0.24 ([10.5281/zenodo.5656903](https://doi.org/10.5281/zenodo.5656903)) (Q4: SCIE: Impact Factor:0.24)
9. Sarita Charde, Malika Mudaliar, Parag Thakur, Shriram Sonawane, Artificial neural network model for prediction of viscoelastic behavior of polycarbonate composites, Journal of Applied Research Technology, 20 (2), (2022) (Q3: Scopus: Impact Factor: NA) (<https://doi.org/10.22201/icat.24486736e.2022.20.2.1101>)
10. Sonawane, S. S., Thakur, P. P., & Paul, R. Study on visco-elastic properties enhancement of MWCNT based polypropylene nanocomposites. Materials Today: Proceedings, 2020, 29, 929-933. (<https://doi.org/10.1016/j.matpr.2020.05.417>) (Scopus)
11. Sonawane, S. S, Thakur, P. P., Paul, R. Study of thermal property enhancement of MWCNT based polypropylene (PP) nanocomposites, Materials Today: Proceedings, 2019, Volume 27(P1), 550-

➤ Research Projects

1. Development of Stable hybrid nanofluid system for CO₂ absorption by SVNIT Surat as Seed Money Grant (Phase-III), from 01.05.2026 to 30.04.2028 (Rs. 10,00,000)

➤ Patents Granted

1. A hybrid nanofluids based drilling fluid and method of preparation thereof, Application number: 202221050711, Date of filling 06/09/2022, Date of Grant: 28/08/2024 (Patent No. 548836)
2. Biological degradation of Sulphur waste from the mill rejects of thermal power plant, Application number: 202221047477, Date of filling 20/08/2022, Date of Grant: 12/06/2025 (Patent No. 567372)
3. Development of Novel Process for micro-reactor-based Extraction of Heavy Antimony using ionic liquid-based Hybrid Nanofluids, Application number: 202321016794, Date of filling 13/03/2023, Date of Grant: 22/05/2026 (Patent No. 590073)

➤ Books Authored

1. Nanofluids: Fundamentals, Applications & Challenges, Emerging Materials and Technologies Series, CRC Press, Taylor and Francis Publications, 2024 (ISBN: 9781032519876)
2. Nanofluids for Efficient Energy Conservation and Process Intensification, CRC Press, Taylor and Francis Publications, 2026 (ISBN 9781041043683)

➤ Book Chapters

1. Parag Thakur, Shriram Sonawane, Chapter 32. Bi-OX-based 2D composites for solar energy harvesting, Bi-OX-based Photocatalysts for Dual Applications, ELSEVIER, 2026, (ISBN: 9780443238611) (<https://doi.org/10.1016/B978-0-443-23861-1.00028-6>)
2. Parag Thakur, Malika Mudaliar, Shriram Sonawane, Muthupandian Ashok Kumar, Chapter 1: Current overview of applications of the hybrid nanofluids, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00001-2>)
3. Parag Thakur, Shriram Sonawane, Dr. Prakash Jadhav, Srinath Suranani, Chapter 3: Overview on Mass transfer enhancement mechanism using the nanofluids, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00003-6>)
4. Parag Thakur, Malika Mudaliar, Shriram Sonawane, Srinath Suranani, Hussain Mohammad, Chapter 4: Overview of the Heat transfer Enhancement mechanism using the nanofluids, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00004-8>)
5. Parag Thakur, Malika Mudaliar, Shriram Sonawane, Irina Potoroko, Ashok Kumar Muthupandian, Chapter 5: Numerical and experimental investigations of nanofluids application solar collectors for the energy optimization, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00005-X>)
6. Aaditi Pargaonkar, Parag Thakur, Shriram Sonawane, Chapter 6: Applications of nanofluid in

- boiling operation for enhancement of the critical heat flux (CHF), *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00006-1>)
7. Parag Thakur, Malika Mudaliar, Shriram Sonawane, , Chapter 10: Applications of the hybrid nanofluids in the CO₂ absorption and desorption processes, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00010-3>)
 8. Vishal Chandane, Ajit Rathod, Parag Thakur, Shriram Sonawane Chapter 11: Applications of hybrid nanofluids in the extraction processes, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00011-5>)
 9. Vishal Chandane, Pradeep kumar Ramteke, Ajit P. Rathod, Parag Thakur, Shriram Sonawane Chapter 12: Applications of hybrid Nanofluids in the Proton Exchange Membrane Fuel Cells, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00012-7>)
 10. Parag Thakur, Malika Mudaliar, Shriram Sonawane, Chapter 13: Challenges of nanofluid applications in the chemical industry, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00013-9>)
 11. Malika Mudaliar, Parag Thakur, Shriram Sonawane, Chapter 15: Applications of the nanofluids for the drilling fluid, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00015-2>)
 12. Malika Mudaliar, Parag Thakur, Shriram Sonawane, Chapter 16: Application of the nanofluids for the oil mobility, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00016-4>)
 13. Uma Sankar Behera, Parag Thakur, Shriram Sonawane, Chapter 17: Nanofluids applications in chemical enhanced oil recovery: current opinion and recent advances, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00017-6>)
 14. Malika Mudaliar, Parag Thakur, Shriram Sonawane, Chapter 18: Investigation of the effects of nanofluids on the hole cleaning efficiency of water-based drilling mud, *Applications of Hybrid Nanofluids in Chemical and Petroleum Industry*, ELSEVIER, 2025 (ISBN: 9780443214516) (<https://doi.org/10.1016/B978-0-443-21451-6.00018-8>)
 15. Parag Thakur, Shriram Sonawane, Introduction to nanofluids, nanofluid applications for advanced thermal solution, ELSEVIERs, 2023, 1-19 (ISBN: 9780443152399) (<https://doi.org/10.1016/B978-0-443-15239-9.00001-1>)
 16. Parag Thakur, Aaditi Pargaonkar, Clara Gongloves, Shriram Sonawane, Synthesis and characterization of Nanofluids, nanofluid applications for advanced thermal solution, ELSEVIERs, 2023, 21-41, (ISBN: 9780443152399) (<https://doi.org/10.1016/B978-0-443-15239-9.00002-3>)
 17. Parag Thakur, Irina Potoroko, Shriram Sonawane, Stability of nanofluids, nanofluid applications for advanced thermal solution, ELSEVIERs, 2023, 43-62, (ISBN: 9780443152399) (<https://doi.org/10.1016/B978-0-443-15239-9.00003-5>)

18. Akash Kumar, Parag Thakur, Malika Manjakuppam, Shriram Sonawane, Promising nanoparticles for water reuse, *Resource Recovery in Drinking waters*, ELSEVIERs, 2023, 129-145 (ISBN: 978-0-323-99344-9) (<https://doi.org/10.1016/B978-0-323-99344-9.00003-7>)
19. Parag Thakur, Manjakuppam Malika, Shriram Sonawane, Energy recovery from industrial wastewaters, *Resource Recovery in Industrial Wastewaters*, ELSEVIERs, 2023, 319-336 (ISBN: 978-0-323-95327-6) (<https://doi.org/10.1016/B978-0-323-95327-6.00009-9>)
20. Manjakuppam Malika, Parag Thakur, Shriram Sonawane, Sulfate/Sulfur recovery from municipal wastewater treatment plants, *Resource Recovery in Municipal Wastewaters*, ELSEVIERs, 2023, 145-164 (ISBN: 978-0-323-95327-6) (<https://doi.org/10.1016/B978-0-323-99348-7.00003-5>)
21. Parag Thakur, Shriram Sonawane, Current overview of nanofluid applications, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 1-26 (ISBN: 978-0-32-390564-0) (<https://doi.org/10.1016/B978-0-323-90564-0.00004-0>)
22. Parag Thakur, Shriram Sonawane, Ratiram Chaudhary, Thermo-physical and optical properties of the nanofluids, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 27-52 (ISBN: 978-0-32-390564-0) (<https://doi.org/10.1016/B978-0-323-90564-0.00006-4>)
23. Dhananjay Singh, S K Patel, P Kumar, D Pal, Parag Thakur, Shriram Sonawane, Experimental investigations of direct absorption solar collectors, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 107-132 (ISBN: 978-0-32-390564-0) (<https://doi.org/10.1016/B978-0-323-90564-0.00011-8>)
24. Parag Thakur, Irina Potoroko, Shriram Sonawane, Numerical and experimental investigations of nanofluid application in car radiators, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 133-162 (ISBN: 978-0-32-390564-0) (<https://doi.org/10.1016/B978-0-323-90564-0.00008-8>)
25. Shriram Sonawane, Parag Thakur, Sparsh Bhaisare, Prakash Jadhav, Mathematical, Numerical and experimental investigations of nanofluids applications in Pool Boiling process, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 163-184 (ISBN: 978-0-32-390564-0) (<https://doi.org/10.1016/B978-0-323-90564-0.00003-9>)
26. Shriram Sonawane, Parag Thakur, Sparsh bhaisare, Prakash Jadhav, Mathematical, Numerical and experimental investigation of nanofluids application in Flow Boiling processes, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 185-204 (ISBN: 978-0-32-390564-0) (<https://doi.org/10.1016/B978-0-323-90564-0.00002-7>)
27. Parag Thakur, Shriram Sonawane, Mathematical and Numerical investigations of CO₂ absorption and desorption process, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 205-226 (ISBN: 978-0-32-390564-0) (<https://doi.org/10.1016/B978-0-323-90564-0.00007-6>)
28. Parag Thakur, Hasan Uslu, Shriram Sonawane, Experimental investigation of CO₂ absorption process using nanofluids, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 227-250 (ISBN: 978-0-32-390564-0) (<https://doi.org/10.1016/B978-0-323-90564-0.00001-5>)
29. ST Yerpude, AK Potbhare, PR Bhilkar, Parag Thakur, Pratiksha Khiratkar, Martin F Desimone, PR Dhongle, Shriram Sonawane, Clara Goncalves, RG Chaudhary, Computational analysis of nanofluids based drug delivery system: preparation, Current Development, and applications of

nanofluids, *Applications of Nanofluids in Chemical and Bio-medical Processing Industry*, ELSEVIER, 2022, 335-364 (ISBN: 978-0-32-390564-0)

(<https://doi.org/10.1016/B978-0-323-90564-0.00014-3>)

30. Shriram Sonawane, Abhijit Gadhe, Parag Thakur, Shirish Hari Sonawane, Hussein A Mohammed, Nano-biotechnology for bacterial modification for wastewater treatment and resource recovery, *Novel Approaches Towards Wastewater Treatment and Resource Recovery Technologies*, ELSEVIERs, 2022, 295-312 (ISBN: 978-0-32-390627-2)
(<https://doi.org/10.1016/B978-0-323-90627-2.00008-3>)
31. Shriram Sonawane, Malika Mudaliar, Parag Thakur, Shirish Sonawane, Carbon Nano tubes (CNT) based hybrid Nanofluids for the waste water treatment plants in the industry, *Novel Approaches Towards Wastewater Treatment and Resource Recovery Technologies*, ELSEVIERs, 2022, 313-324 (ISBN: 978-0-32-390627-2) (<https://doi.org/10.1016/B978-0-323-90627-2.00021-6>)
32. Parag Thakur, SS Sonawane, SH Sonawane, Nano-particle-enhanced ionic liquids (NEIL) for the waste water treatment, *Novel Approaches Towards Wastewater Treatment and Resource Recovery Technologies*, ELSEVIERs, 2022, 325-338 (ISBN: 978-0-32-390627-2)
(<https://doi.org/10.1016/B978-0-323-90627-2.00020-4>)
33. Parag Thakur, Shriram Sonawane, Nanomaterials for membrane synthesis: Introduction, mechanism and Challenges for wastewater treatment, *Handbook of Nanomaterials for Wastewater Treatment: Fundamentals, Current status and Scale-up Challenges*, ELSEVIER, 537-553 (ISBN: 978-0-12-821496-1) (<https://doi.org/10.1016/B978-0-12-821496-1.00009-X>)
34. Parag Thakur, Sonawane, S. S., Sonawane, S. H., & Bhanvase, B. A., Nanofluids-based delivery system, encapsulation of nanoparticles for stability to make stable nanofluids. *Encapsulation of Active Molecules and Their Delivery System*, ELSEVIER, 141-152 (ISBN No. 978-0-12-819363-1)
(<https://doi.org/10.1016/C2018-0-05369-4>)
35. Parag Thakur, Nirvik Sen, Sameer Ekhande, R. Sirsam, K.K. Singh, S. Mukhopadhyaya, K.T. Shenoy, Reactive Stripping & Precipitation of Uranium in microreactors, *International Nuclear Information System (INIS)*, 2018, Volume 49, Issue 37. (RN:49072564)

➤ **Guest Editor**

1. Current Pharmaceutical Biotechnology, Bentham Science Publications, Impact Factor: 2.4 (July 2025-June 2026)
2. Water Environment Research, Wiley Publications, Impact Factor: 1.9

➤ **Reviewer of Journals**

1. Separation Science & Technology (Taylor & Francis Publication; Impact Factor: 1.2)
2. Current Pharmaceutical Biotechnology (Impact Factor: 2.4)
3. Current Nanomedicine (Impact Factor: 1.6)

➤ **Certification & Memberships**

1. Certified Auditor, Gujarat Pollution Control Board, Government of Gujarat
2. Member, Institution of Engineers

➤ **Industry Research Collaboration / Consultancy**

1. Green Gene Enviro Protection and Infrastructure Ltd, Surat (GGEPIL). - August 2026–Present.

SECTION C: Outreach Activities



➤ Conference Activities

Conference Organized:

1. Secretary, 2nd Edition of International Conference on Advances in Sustainable Research for Energy and Environmental Management (ASREEM-2026), 15-17 May 2026, SVNIT, Surat

Session Chair:

2. International Conference on Advances in Sustainable Technologies and Industrial Innovations (ICASTII-2026) scheduled at Jawaharlal Darda Institute of Engineering and Technology, Yavatmal (24–25 March 2026)
3. ASTHRA-A National level Technical Symposium at Department of Chemical Engg, BVRIT, Narsapur, Hyderabad, Telangana (17-18 Oct, 2025)
4. International Conference on Education 5.0-Role of Institution, Industry and Society (ERIIS-2022) at NIT, Warangal (14-15 October 2022)
5. 18th Annual Session of Chemical Engineering Students Congress (SCHEMCON 2022) at NIT, Warangal (23rd to 24th September 2022)

Invited Talks and Keynote Lectures:

6. “Artificial Intelligence and Machine Learning”, National Conference on Cutting edge Research and Innovation 2026 (NCCERI-2026) organized by Bharati Vidyapeeth, Pune (27 and 28th March 2026)
7. ‘Data Driven hydrogen production systems’, International conference on Advances in Chemical Technology & Allied Sciences (ACTAS-2026)’, University Institute of Chemical Technology, Jalgaon. Duration: 6-7 January 2026 (Lecture Date: 6 January 2026)
8. ‘Data Calculation, manipulation & Visualization in Python’, 3rd Edition of One-week Short Term Training Program (STTP) on ‘Simulation Techniques for Engg. Problems’ at SVNIT, Surat (1-5 Nov. 2025)
9. International Conference on Nano Structured Materials and Nanocomposites (ICN 2025) at Mahatma Gandhi University, Kottayam, Kerala, India (12-14 September 2025)
10. ‘Nanofluids based hydrogen storage systems’, AICTE-QIP-PG Programme on “Advanced Metallic Alloys for Energy Storage Applications”, Department of Mechanical Engineering, SVNIT, Surat. Duration :16th June 2025 to 31st December 2025 (Lecture Date: 12 July 2025)
11. ‘Data Science for Instrumentation Techniques’, 2nd Edition of One-week Short Term Training Program on ‘Instrumentation Techniques for the environmental remediation and hands-on Training (ITER-2025) organized by SVNIT, Surat. (21 May 2025)
12. ‘Nanofluids: Fundamentals, Applications and Challenges’, One-week Short Term Training Program (STTP) on Nanotechnology: Trends and Application organized by Jawaharlal Darda Institute of Engg. And Technology, Yavatmal (14 January 2025)
13. ‘Introduction to SCILAB’, 1st Edition of 6-days Short Term Training Program (STTP) on ‘Introduction to Chemical Engineering Software (IChES-25) organized by SVNIT, Surat. Lecture Date: 12 January 2025
14. ‘Application of nanotechnology in Wastewater treatment’, 1st Edition of One-week Short Term

Training Program on 'Instrumentation Techniques for the environmental remediation and hands-on Training (ITER-2024) organized by SVNIT, Surat. (15 May 2024)

Conference Papers Presented:

15. Parag Thakur, Application of nanofluids for the PVT solar panels, 16th International conference on advancement in petrochemical sector and empowering sustainable development leading to Viksit Bharat-APM 2025 organized by Central Institute of Petrochemical engineering & Technology (CIPET): IPT-Lucknow 8th-10th March, 2025(Received Best oral presentation award)
16. Parag Thakur, Experimental analysis of fly ash based nanofluids for CO₂ absorption, International Conference on emerging materials trends in chemistry to revolutionize Indian chemical industries for Viksit Bharat@2047 organized at JECRC University, Jaipur, 19th-21st December, 2024
17. Parag Thakur, Fly based nanofluids for thermal applications, International Conference on Water & Sustainable Development 2024 (ICWSD 2024) organized by VNIT, Nagpur, 8-9 November 2024
18. Parag Thakur, Experimental and numerical analysis of CO₂ absorption processes using fly ash based nanofluids, 15th International conference on advancement in polymeric materials-APM 2025 organized by Central Institute of Petrochemical engineering & Technology (CIPET): IPT-Ahmedabad, 14th-16th March, 2024
19. Parag Thakur, Shriram Sonawane, Recent trends of CO₂ absorption process, 6th Conference by Asia-Oceania Sono-chemical Society (AOSS-23), NIT, Warangal, 28-30 September, 2023 **(Received Best oral presentation award)**
20. Parag Thakur, Shriram Sonawane, Experimental and numerical analysis of CO₂ absorption processes using fly ash based nanofluids, CHEM-CONFLUX-22, NIT, Prayagraj, 14-16 April, 2022 **(Received Best oral presentation award)**
21. Synthesis of BNNP+Mxene/Polycarbonate nanocomposite using twin-screw extrusion reactor for thermal property enhancement, International Chemical Engineering Conference 2021 (ICheEC 2021), September 17-19, 2021, NIT, Jalandhar
22. Parag Thakur, Pooja Pradip, Shriram Sonawane, Development of efficient CO₂ absorption processes using MWCNT based nanofluids, 2nd International Conference on Chemical, Bio & Environmental Engineering (CHEMBIOEN-2021), August 20-22, 2021, NIT, Jalandhar
23. Parag Thakur, Shriram Sonawane, Performance assessment of the CNT deposited surfaces for the pool boiling application, International Conference on Advances in Sustainable Research for Energy and Environmental Management (ASREEM-2021), 6-8 August 2021, SVNIT, Surat **(Received Best oral presentation award)**
24. Parag Thakur, Tushar Khapne, Shriram Sonawane, Synthesis of BNNP+Mxene/Polycarbonate nanocomposite using twin-screw extrusion reactor for thermal property enhancement, International Conference on Advances in Sustainable Research for Energy and Environmental Management (ASREEM-2021), 6-8 August 2021, SVNIT, Surat.
25. Parag Thakur, Shriram Sonawane, Review on the Application of Nanofluids for the Environmental Remediation, International Conference on Advances in Sustainable Research for Energy and Environmental Management (ASREEM-2021), 6-8 August 2021, SVNIT, Surat.
26. Synthesis of BNNP+Mxene/Polycarbonate nanocomposite using twin-screw extrusion reactor for thermal property enhancement, International Conference on Reaction Engineering (ICRE-

2021) 7th- 8thMay, 2021, NIT, Raipur

27. Parag Thakur, Shriram Sonawane, Photo-thermal Removal of CO₂ from MWCNT/HMDA nanofluid, The Ninth DAE-BRNS Biennial Symposium on “Emerging Trends in Separation Science and Technology (SESTEC-2020), 22-26 March 2021, BARC, Mumbai
28. Comparative study on the thermal property and haze enhancement using boron nitride nanosheets and MXene nanoparticles in a polycarbonate matrix, MHRD, GoI, TEQIP-III Sponsored International E-Conference on Recent Transformations in Chemical & Textile Technology, 24-26 August 2020.
29. Parag Thakur, Shriram Sonawane, Numeric and experimental study of the car radiator performance, CHEM-CONFLUX-20, 14-16 February 2020, NIT, Prayagraj
30. Praneet Lokhare, Rahul Bhad, Parag Thakur, Shriram Sonawane, Numeric and experimental investigation of cobalt oxide-based nanofluids in direct absorption solar collector, International Conference on Multifunctional and hybrid materials for chemical process, energy, environment and medical applications (ICMHCEE 2019), 9-11 September 2019, NIT, Trichy, Tamil Nadu
31. Shriram Sonawane, Parag Thakur, Ritesh Paul, “Study of thermal property enhancement of MWCNT based polypropylene (PP) Nanocomposites”, First International Conference on Recent Advances in Materials and Manufacturing (ICRAMM 2019), 12-14 September 2019, KLE Dr. M.S. Sheshgiri COET, Belagavi, Karnataka.
32. Shriram Sonawane, Parag Thakur, Ritesh Paul, Study of Visco-Elastic property enhancement of MWCNT based polypropylene (PP) Nanocomposites, 11th National Conference on Solid State Chemistry and Allied Areas (NCSCA-2019), 20-21 December, 2019, S.K. Porwal College, Nagpur
33. Parag Thakur, Mayur Darekar, K.K. Singh, S. Mukhopadhyaya, K. T. Shenoy, Numerical Simulation of Liquid-liquid two-phase flow at different Microfluidics junctions, The Eighth DAE-BRNS Biennial Symposium on “Emerging Trends in Separation Science and Technology (SESTEC-2018), May 23-26, 2018, BITS, Pilani (Goa).
34. Parag Thakur, Nirvik Sen, K.K. Singh, S. Mukhopadhyaya, K.T. Shenoy, Reactive Stripping & Precipitation of Uranium in microreactors, The Eighth DAE-BRNS Biennial Symposium on “Emerging Trends in Separation Science and Technology (SESTEC-2018), May 23-26, 2018, BITS, Pilani (Goa).

Conference Attended:

35. International Conference on Multifunctional Electronic Materials and Processing (MEMP-2021) conducted by C-MET Pune on digital platform during 8-10th March 2021.
36. International Conference (online mode) on Chemical Science for Drug Discovery & Therapy 2020, during 22-26 July, 2020, Organized by Department of Chemistry VNIT Nagpur, Nagpur, India.
37. 1 Day online conference in intellectual property and startups on 20th January 2024 event organized by turnip Innovation festivals 2024

➤ Short Term Training Program/Workshops

STTP/Workshop Organized:

1. Workshop on ‘Introduction to Python’ in collaboration with IIT, Bombay, (22 - 24 Dec ,2025)
2. 3rd Edition of One-week Short Term Training Program (STTP) on ‘Simulation Techniques for Engg. Problems’ at SVNIT, Surat (1-5 Nov. 2025)
3. One-week Short Term Training Program (STTP) on ‘Introduction to Standards’ Sponsored by

Bureau of India Standards at SVNIT, Surat (2nd, 4th-5th, 10th-11th Oct, 2025)

4. 3rd National NITs Training & Placement (T&P) Conclave, SVNIT, Surat (20 June 2025)
5. 2nd Edition of One-week Short Term Training Program on 'Instrumentation Techniques for the environmental remediation and hands-on Training (ITER-2025) at SVNIT, Surat from 19-23 May 2025
6. 1st Edition of 6-days Short Term Training Program (STTP) on 'Introduction to Chemical Engineering Softwares (IChES-25) at SVNIT, Surat on 11-12 January, 18-19 January, 25 January and 1st February 2025
7. 1st Edition of One-week Short Term Training Program on 'Instrumentation Techniques for the environmental remediation and hands-on Training' (ITER-2024) at SVNIT, Surat from 13-17 May 2024

STTP/Workshop Attended:

8. 4 days Faculty development Program by art of living from 24 August to 27 August 2026 at SVNIT, Surat.
9. One day workshop on 'Power Aware UAV Networks' on 29 June 2026 at SVNIT, Surat
10. One day lectures on 'Design Education: Philosophy and Future Scope' by Prof. Amit Ray on 15 April 2026 at SVNIT, Surat
11. Faculty Development Program on 'Innovation, Problem Solving and human Centered Design' on 24 April 2026 organized by MasterCard, Pune
12. One day Start-up awareness program/sensitization program for Start-up on 23rd march 2026 at SVNIT, Surat
13. One day program 'University Relations Day' on 13 February 2026 organized by MasterCard, Pune
14. Online training programme titled "Leadership, Motivation, Decision Making, and SWOT Analysis", from 22 to 26 December 2025 organized by NITTTR, Bhopal
15. Two days Pedagogy Workshop on Design Thinking and Entrepreneurship under Capacity Building Program on Design and Entrepreneurship (CBDE) under the Malaviya Mission Teacher Training Programme (MMTTP) at SVNIT, Surat from 8th July 2025 to 9th July 2025
16. 1 week workshop on Empowering educators: student centric teaching pedagogy at SVNIT, Surat from 9 June to 13 June 2025
17. One day workshop on REINVENT at SVNIT, Surat on 3rd May 2025
18. Two days' workshop on pedagogy to promote design and entrepreneurship under Capacity Building Program on Design and Entrepreneurship (CBDE) under the Malaviya Mission Teacher Training Programme (MMTTP) at ABV-IIITM, Gwalior from 6th march 2025 to 7th March 2025
19. 1 day Health Awareness Program organized by SVNIT, Surat on 22 Feb 2025
20. 4-day faculty development programme on 'Recent Pedagogies for Better Learning' organized by SVNIT, Surat from 22-25 July 2024
21. One week faculty development program on Education and Institutional development (EDID-2024) organized by SVNIT, Surat from 15th-21st May, 2024
22. 1 Day workshop on Hindi workshop on 29 April 2024 at SVNIT, Surat
23. 1 Week faculty Development program on cyber-attacks and defense conducted by C-DAC, Noida from 2nd January 2024 to 6th January 2024
24. 1 week GIAN (Global initiative of Academic network) on "Greener and cleaner ultrasonic process for the production of nanomaterials and nano-pharmaceuticals" organized by NIT, Warangal, 27

June-2 July, 2022 (Foreign Faculty: Manickam Sivakumar, Brunei)

25. 1 week GIAN (Global initiative of Academic network) on "Green Processing & Synthesis" organized by VNIT, Nagpur, 2-6 May, 2022. (Foreign Faculty: Prof. Ashok Kumar Muthupandian)
26. National Level Webinar on 'Writing and Evaluation of Scientific Research Articles' jointly organized by Research Promotion Cell, Seth Kesarimal Porwal College of Arts, Science and Commerce, Kamptee & Department of Chemistry, Taywade College, Mahadula-Koradi, 31st July 2021
27. AICTE ATAL sponsored 1 week workshop on Nanotechnology for the sustainable development and green processes, 24-28 August, 2020, VNIT, Nagpur (Secured A+ Grade)
28. 1 day workshop on why manuscript get rejected, NIT, Jalandhar, 27 August 2020
29. 63rd ISRO-IIRS Outreach Program on "Remote Sensing Applications in Agricultural Water Management" August 3-7, 2020 (Secured A grade)
30. Webinar "Opportunities for Chemist and Chemical Engineers in Chemical and Allied Industries, VJTI, Mumbai, 18 July 2020
31. 1-week STTP on CFD with OpenFOAM, July 23-28 2020, VNIT, Nagpur
32. 1 Week FDP on Pedagogy of Scientific Writing, Reporting and Scholarly Networks, Organized by Feroze Gandhi Institute of Engineering & Technology, Rae-Bareilly During June 19-23, 2020
33. 1 day webinar on Understanding open educational resources offered by commonwealth of learning, Canada, 23 June 2020
34. 7 Week online course on LATEX organized by IIT Bombay, 24 September- 11 November 2019
35. Two-day Training Session on TGA and DSC and Two-day Training on DLS organized by VNIT, Nagpur, 3-6 April 2019
36. Two days Training Session on DMA & Mechanical Hybrid Rheometer organized by TA Waters, Bangalore, 12-13 March 2019
37. One week GIAN Workshop on Process Intensification organized by NIT, Warangal, 7-11 January 2019 (Foreign Faculty: Andrzej Górak)
38. Three days' Workshop on Python at organized by VNIT, Nagpur, 10-12 August 2018
39. One-week Short Term Training Program (STTP) on Advanced Industrial Waste Management Techniques at University Institute of Chemical Technology, North Maharashtra University, Jalgaon, Maharashtra, 14-19 May 2018
40. Two-week Short Term Training Program (STTP) on Process Intensification: Fundamentals to Applications at University Institute of Chemical Technology, North Maharashtra University, Jalgaon, Maharashtra, 26 December -4 January 2017

SECTION D: Administrative Activities



➤ Institute level

Sr. No.	Section	Assigned Responsibility
1.	Hostel Section	1. Warden: Bhabha Bhawan-Boys Hostel, SVNIT, Surat (From 4 Oct. 2025-) 2. Co-Chairperson, Mindbend-2026, A Technical Event of SVNIT (from 12 Jan. 2026 to Present) 3. Co-Chairperson, Chemical Engineering Society (Student Chapter of AIChE), SVNIT (from 12 Jan. 2026 to Present) 4. Member: Institute Level Anti-Ragging Committee (2026-27) 5. Member: Stock Verification Committee, Narmad Bhawan, ABV Bhawan (2025-26) 6. Member: Mess Inspection Committee, ABV Bhawan (2025-26) 7. Member: Hostel Level Anti-Ragging Committee, Boys hostel (Gajjar Bhawan) (2025-26, 2024-25) 8. Member: Invitation & Sponsorship Committee, Kashish-2025 (8-9, Oct, 2025) 9. Member: Organizing Committee, Sparsh-2026 (11-13 April, 2026)
2.	Academic Section	1. Member, Committee for National Institutional Ranking Framework (NIRF) (2025-26, 2026-27) 2. Member Secretary: Committee for Orientation Program for First year UG, PG & PhD students (2026, 2025, 2024) 3. Member: B. Tech and M. Sc First Year Admission committee through JoSAA/CSAB (2025-26, 2026-27) 4. Member: B Tech I, Fee Remission Committee (2023-24, 2024-25, 2025-26) 5. Member: Disciplinary Committee and Registration Sub-committee, • 21st Convocation, SVNIT, Surat (14th Oct 2025) • 22nd Convocation, SVNIT, Surat (18th August 2026) 6. Member: PhD Thesis Evaluation Committee of following students: • Mr. Gajera Jeet Bhovanbhai (D20MA007), Date: 10/04/2024 • Ms. Rashmita Behera (DS19CH001), Date: 06/06/2025 • Ms. Shobha Rawat (DS19CE012), Date: 06/06/2025 • Ms. Nidhi (D22MA004), Date 08/07/2026
3.	Research & Consultancy	1. Single Point of Contact, Biotechnology Theme, Prime Minister Research Chair Scheme, 2026
4.	Purchase Section	1. Member: Committee for Tender of networking work at various departments & Sections of Institute (2025-26)
5.	Establishment	1. Member: Document verification Committee, Recruitment of professor

	Section	and assistant professor-Gr-I and Upgradation of internal faculty members from Pay level 10 to Pay level 11 for Department of electrical engineering, (3rd August 2025) 2. Member, Scrutiny Committee, Recruitment of Superintendent 2025, SVNIT, Surat 3. Member: Committee of Paper Checking for - Assistant Professor-Level 10 (16-07-2026), - Assistant Registrar (16-07-2026)
6.	Other	1. Mentor: Capacity Building on Design and Entrepreneurship (CBDE), 2025-26 2. Invigilation Duties: - NEET 25 (4th May 2025) at SVNIT, Surat - NEET 26 (3rd May, 2026 and 21st June 2026) at SVNIT, Surat - Junior Assistant Exam, (15-17 February 2026) at SVNIT, Surat 3. Presiding officer: - Surat Municipal Corporation Elections (26 April 2026) - Gujarat Legislative Assembly Elections (7 May 2024)

➤ Department level

Sr. No.	Role	Assigned Responsibility
1.	Lab Co-In charge	CAD lab (New building and old building)
2.	Department Coordinator	1. Career Development Cell (formally Training & Placement Cell) (19 April 2024-Present) 2. Internship of UG & PG students (From 12/01/24 to Present) 3. Departmental website/social media update, Media Cell (August 2024-Present) 4. Orientation Program of BTech 1st year (16-18 August 2024), MTech 1st year and PhD 1st year (30-31 August 2024)
3.	Member-Secretary	1. Committee for Higher Studies/Career Counselling (from 3/10/24 to Present) 2. Committee for Accreditation & Academic Audit (from 3/10/24 to Present) 3. Committee for Utilization of Various Open & Free Source Software (from 3/10/24 to Present) 4. Committee to review the admission rules of PhD program (2025-26)
4.	Co-Chairperson	BIS Standards Club, (from 26/12/23 to Present)
5.	Member	1. Selection Committee, Teaching Assistant (2025-26) (16-06-2025; 18-08-2025; 21-01-2026) 2. Member: PhD Admission Committee (Dec 2023, June 2024, Jan 2025, July 2025, Jan 2026, July 2026)

		3. Committee for Swachhotsav (17/09/2025 to 2/10/2025)
6.	Document Verification Committee Member	1. MTech Admission (Institute Spot Round), 2024-25, 2025-26 2. PhD admission (QIP) 2024-25, 2025-26 3. PhD admission: 2026-27, Autumn Semester (Phase-III), Date: 21-07-2026; AY 2023-24; Spring Semester, Date: 12-12-2023
7.	Application Scrutiny Committee Member	1. MTech Admission (Institute Spot Round), 2024-25, 2025-26, 2026-27 2. PhD admission, 2026-27, Autumn Semester (Phase-III) 3. PhD admission, 2024-25, Autumn Semester, (Phase-II) 4. PhD admission, 2025-26, Spring Semester, (Phase-II) 5. PhD admission, 2025-26, Autumn Semester, (Phase-II) 6. Summer Internship 2026
8.	PhD Examiner	1. Ms. Monali Chhatbar (D18CH006) 2. Ms. Rashmita Patel (D23CH001) 3. Mr. Arth Gandhi (D23CH002) 4. Mr. Abhishek Mehta (D23CH003) 5. Mr. Amit Kumar (D21CH003) 6. Ms. Priti Sakhare (D24CH002) 7. Mr. Bhargav Shukla (D24CH005)
9.	MTech Examiner	8. Mr. Jay Pandya (P22CH009) 9. Mr. Shivam Modi (P22CH001)
10	Purchase Committee member	1. FO-MD setup fabrication (DoChE/181/2024) (Approximate cost: Rs. 2,48,980) 2. Liqui-Cel™ EXF 2.5X8 Series membrane contactor GS01, Polypropylene X50 (DoChE/562/2024) (Approximate cost: Rs. 4,90,000) 3. Osmometer (DoChE/113/2024) (Approximate cost: Rs. 7,00,000) 4. Probe-type Sonicator (Approximate cost: 6,66,000) 5. 10 Desktop PCs (Approximate cost: 5,00,000)

